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Growing Rice in Kano State, Northwestern, Nigeria

A.Y. Kamara, L.O. Omoigui, K.T. Aliyu,
A.I. Tofa, S. Miko, R. Solomon,
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March 2022

Published by the International Institute of Tropical Agriculture (IITA)
Ibadan, Nigeria

International address:
IITA, Grosvenor House,
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Croydon CR0 9XP, UK

Headquarters:
PMB 5320, Oyo Road
Ibadan, Oyo State

ISBN 978-978-131-401-8

Citation: Kamara, A.Y., L.O. Omoigui, K.T. Aliyu, A.I. Tofa, S. Miko, R. Solomon, A.H. Kmata, and A. Raba. 2022. Growing Rice in Kano State, Northwestern, Nigeria. pp. 22

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Cover: Paddy rice at maturity stage.



Acknowledgements

This handbook is produced to help farmers in Kano State use improved varieties and good agronomic practices to increase rice productivity in the State. This guide draws its lessons from the work and experience of IITA and partners in Research and Development on rice-based systems in northern Nigeria. We believe the handbook will enrich the knowledge of the extension personnel and smallholder farmers to use agronomic technologies to address the production challenges and increase the productivity of rice in the State.

The publication was funded by the Islamic Development Bank, Lives and Livelihood Funds and the Kano State Governments through funds provided to the Sasakawa Africa Association for the execution of the Kano State Agro-pastoral Development Project (KSADP).

We acknowledge the contribution of our partners in the development of this handbook. The availability of extensive research results and the feedback from farmers and partners have helped develop this handbook.

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Introduction

Rice is one of the major staple foods in Nigeria, consumed across all geopolitical zones and socioeconomic classes (Kamai et al. 2020). Rice consumption is increasing rapidly in Nigeria because of a shift in consumer preference towards rice, increasing population growth, increased income levels, and rapid urbanization. In Kano State, rice is the fourth most important cereal crop grown by farmers after sorghum, millet, and maize. Though rice production is not as high as that of the other cereal crops, Kano is among the major rice producing states in northern Nigeria. Others are Kebbi, Borno, and Kaduna states. Though rice is grown in the upland areas, most rice is grown on flooded plains called fadamas where most of the crop is irrigated.

Yields of rice are, however, low in Kano State. Most farmers producing rice rely on traditional technology with few improved inputs. The average rice yield in Kano State is very low (1.5 t/ha) compared to a yield of 6–8 t/ha reported on research plots. To increase rice production and productivity in the State, farmers would have to adopt improved varieties and good agronomic practices.

Constraints to rice production in Kano State

Drought

Drought is a major constraint to rice production in northern Nigeria because rice requires a lot of water for optimum growth and yield. Rice requires about 1200 mm to 1600 mm of rainfall evenly distributed throughout its growing period. This volume of rainfall cannot be obtained in Kano State where the Sudan savanna zone dominates with less rainfall than other zones. Intermittent drought is also common in Kano State. The severity of drought is higher in the uplands than in the lowlands (fadamas). Rice is therefore mostly grown in the fadamas under rainfed and irrigation conditions.

Rice varieties recommended for Kano State are therefore early and medium maturing and are either tolerant to or can grow fast to escape drought.

Poor soil fertility

Like other parts of northern Nigeria, soil fertility in Kano State is poor due to increased pressure on land resources arising from increased population pressure. There is also a widespread deficiency of nutrients across the major rice producing clusters

in Kano State (Yusuf et al. 2020). In addition to the inherent deficiency of nutrients, salinity and sodicity are seemingly reducing yield due to continuous irrigation with salty water.

Pest attack

One of the major reasons for the low rice yields in the Nigeria savannas is the attack by insect pests. The rice plant is an ideal host for insect pests, root-feeders, stem borers, leaf feeders, and grain feeders. Pests, especially birds, African rice gall midge, and *Striga* attacks, are major constraints to rice production in northern Nigeria. In LGAs with dams (Kura, Garun Malam, Bebeji, Bagwai, Danbatta), attacks by Quelea birds can cause 80% damage to rice.

Important steps in growing rice

Site selection

Rice can be grown in two main ecologies in Kano State:

Lowland areas: These are low lands at the edges of flooded fadamas (inland valleys) and irrigation schemes where water is available for four and a half to five months. In some areas, water may be available for more than five months.

Upland areas: These are well-drained areas with good soil and rainfall of over 700 mm.

Recommended rice varieties

Because of the short growing season prevailing in Kano State, rice varieties recommended for cultivation are mostly early maturing except where irrigation facilities are available. The recommended rice varieties for Kano State are presented in Table 1.

Table 1. Some rice varieties recommended for Kano State.

Variety	Habitat	Characteristics						
		Adaptation	Tillering capacity	Potential yield (t/ha)	Days to maturity (days)	Plant height (cm)	Stem base coloration	Ligule type/ length
FARO 44	Irrigated lowland	Shallow swamp/ Fadama	High	7	95–110	100–115	Green	Long grain
FARO 52	Irrigated lowland	Shallow swamp/ Fadama	High	6	125–135	115–120	Straw	Long grain
FARO 61	Lowland	Shallow swamp/ Fadama	High	5	100–115	90–100	Straw	Long grain
FARO 59	Upland	Rainfed upland	Medium	3	95–100	100–120	Light purple	Intermediate
GAWAL R1	Irrigated lowland	Rainfed upland	High	10	90–105	100–115	Green	Long grain

Source: GoSeed (Business Incubation Platform, IITA).

For access to seeds, please contact the Kano State Agricultural and Rural Development Authority (KNARDA), Sasakawa Africa Association, seed companies and the Africa Rice Center at Ibadan.

Land preparation

Proper land preparation is necessary for rice production to minimize competition with weeds. Two harrowings provide sufficient tilth for rice growth. For irrigated land where transplanting is required, the land should be adequately puddled before transplanting rice.



Plate 1. Land preparation for rice production (a) upland and (b) lowland.

Seed rate for direct seeding

Lowland rice: Use 50–60 kg/ha of seeds.
Upland rice: About 40–50 kg of seeds are required to plant a hectare when sowing by dibbling.



Plate 2. A well-established rice field.

Time of sowing

The time for sowing rice depends on the type of land, production season (irrigated or rainfed), whether it is lowland or upland, and the agroecology where the farm is situated. The recommended time for sowing rice is indicated in Table 2. However, the actual timing of sowing should be adjusted in accordance with the time of the establishment of the rains.

Table 2. Time to plant rice in Kano State for rainy season production.

Ecology	Time of planting
Southern Guinea savanna	Between weeks 1 and 2 in June
Northern Guinea savanna	Between weeks 1 and 2 in June
Sudan savanna	Between week 4 in June and week 1 in July

Lowland rice: Lowland rice can be direct-seeded or transplanted. For direct seeding, plant from late May or early June, depending on the time of rain establishment. As much as possible, adopt the recommended sowing date for your area as indicated in Table 2. For dry season irrigated rice, planting should be done between early February and mid-March

Nursing and transplanting rice

Kano State lowland rice is either direct-seeded or transplanted. Before transplanting, rice seeds should be nursed for 21 days. To establish a nursery, select a good site with well-drained fertile soil and conveniently located in an area close to the main rice field. Prepare the seed bed at least seven days but preferably 14 days before the seeds are sown. Soak the seeds in water for 24 hours and spread them on the floor for 48 hours to sprout using a polythene sheet. Spread the sprouted seeds uniformly on a puddled nursery bed. Use a seed rate of 80–100 kg/ha. Drain the excess water from the nursery bed for about a week and then flood to 2–3 cm depth. Apply 50 g/m2 of NPK 15-15-15 fertilizer. About five days after sowing seeds, the nursery beds should be kept flooded at 2–3 cm depth and gradually increased to 5 cm to keep the beds weed-free. In Kano State, transplant rainfed rice in late June–early July when the rains have established. For irrigated rice, transplant in February when temperature begins to rise after the harmattan period. Transplant seedlings to the well-puddled main fields between 14 and 21 days after sowing. Transplant at a rate of 2–3 seedlings per hill, to a depth of 3–4 cm, and at a spacing of 30 cm × 30 cm (best for late-maturing cultivars), or 20 × 20 cm when soil is fertile or fertilizer is applied.

Spacing

Lowland rice: Sow rice seeds by drilling in rows at a spacing of 20 cm or 30 cm apart. For transplanting, transplant seedlings at a rate of 2–3 seedlings per hill, to a depth of 3–4 cm, and at a spacing

of 30 × 30 cm (best for late-maturing cultivars), or 20 × 20 cm when the soil is fertile or sufficient fertilizer is available.

Upland rice: Dibble 5–6 seeds at a spacing of 20 × 20 cm or 30 × 30 cm and later thin to three to four seedlings per stand at two to three weeks after sowing.

Fertilizer

The following recommendations for lowland rice are similar to those for irrigated rice schemes and fadama areas where there are proper water control and distribution. The water level on the field at the time of fertilizer application must be maintained at 3–5 cm to ensure the efficient use of the applied fertilizer. If the water on the field is more than 10 cm, it will cause a loss of nitrogen fertilizer through volatilization; therefore, drain the field before fertilizer application.

Lowland rice: Apply 60–80 kg of nitrogen and 13 kg of phosphorus (P)/ha (i.e., 30 kg P205/ha) and 25 kg potassium (K)/ha (i.e., 30 kg K20/ha). Apply NPK to supply the full P and K and the first dose of N. Apply four (50 kg) bags of NPK 15: 15:15 at sowing and about two bags of urea at 6–8 weeks after sowing.

Upland rice: The application of 60–80 kg/ha of nitrogen and 30 kg/ha each of phosphorus and potassium (50–30–30) is recommended. This is about four bags of NPK 15:15:15 applied at sowing and two bags of urea applied 6–7 weeks after sowing. Incorporate (bury) the fertilizer in the soil properly to avoid losses. Top dressing of N fertilizer should be applied between the rows and buried.

Weed control

A number of methods can be combined to control weeds in rice. Examples of such methods are fallowing, land preparation, a competitive rice variety, water control, hand weeding, and herbicides.

Hand weeding

Rice fields should be weeded regularly, especially during the early stages of growth. For lowland rice, maintain the field bunds, which are essential for good water control. Maintain an even water depth of about 10 cm all over the fields except when fertilizer is applied.



Plate 3. Manual weeding on rice field.

Weeding should be carried out twice, at 2–3 weeks and 5–6 weeks after sowing. Pull out weeds by hand or with a small hand hoe; heap the weeds outside the field.

Hand weeding can, however, be relatively ineffective, particularly in controlling many of the perennial weeds (e.g., *Cyperus spp.*), which have underground tubers and rhizomes from which they can rapidly reestablish.



Plate 4. A farmer spraying his rice field.

Use of herbicides

Various weeds are associated with the rice crop; it is advisable to use appropriate herbicides that kill various weeds as recommended (Table 3).

Non-selective herbicides, such as glyphosate and glufosinate-ammonium, are sometimes used during land preparation to reduce the problem of perennial weeds. Such weeds are *Cyperus* spp. and *Oryza longistaminata*, which are difficult to control with most herbicides selective to rice.

Table 3. Recommended herbicides for upland and lowland rice production.

Treatment formulation	Commercial name	Rate	Time of application	Remarks
Pendimethalin (455 g/L)	Stomp CS	2.5–3 L/ha	Preemergence	Apply within 2–3 days of planting (for upland and lowland rice). For annual grass and small-seeded broadleaf weed control.
Selective herbicide				
Propanil + 2,4-D amine (360 + 200 g/L) EC	Vespanil plus	1.25–2.5 L/ha	Postemergence	Applied at 1–4 leaves weed stage, irrespective of the stage of the rice or 14–21 days after seeding or transplanting. Selective to rice with broad-spectrum weed control.
Bispyribac-sodium 100/L SC	Rice hope	1–1.5 L/ha	Postemergence	Applied at 2–6 leaves weed stage, irrespective of the stage of the rice or 14–21 days after seeding or transplanting. Selective to rice with broad-spectrum weed control.
Pritilachlor + Pyribenzoxim	Solito 320EC	500 ml/L/ha	Early postemergence	Applied at 2–4 leaves weed stage, irrespective of the stage of the rice or 14–21 days after seeding or transplanting. Selective to rice with broad-spectrum weed control.

‡DAS = Days after sowing, DAT = Days after transplanting
One L/ha is approximately 100 ml/15 L sprayer; thus, 3 L/ha will imply approximately 300 ml/15 L sprayer. A peak milk tin standard size is about 150 ml.

Always read the herbicide label for appropriate instructions before use. Wear protective equipment when applying herbicides.

Pest and disease control

Pests

Rice is less affected by field and storage pests than other cereals grown in northern Nigeria. The major pests of rice are borers and armyworms.

Borers

These are among the most destructive groups of insects that attack rice. They attack the crop from seedling to maturity. The damage caused by various borer species is identical, i.e., the borer cuts off the growing part of the plants from the base, feeds inside the stem, and causes the plant or tiller to die. Borer attack during the flowering stage usually results in empty panicles, i.e., "white heads."

Cultural control: After harvest, burn all stubble from the previous crop. This will destroy the semi-active resting stages of the borers that normally inhabit the stubble. Alternatively, the infested rice field could be flooded after harvest for a week to completely submerge the rice stubble, thereby drowning the borers.

Chemical control: Lamdacyhalothrin at the rate of 1 L/ha. Chemical control measures should be implemented when about 20% of the field is infested.

Armyworms and termites

Upland rice is also commonly attacked by armyworms and termites (root-feeder), especially when the rain stops at the beginning of the season. Application of Cypermethrin or Lamdacyhalothrin at 1 L/ha is effective against armyworms.

Birds

Bird attacks are serious problems in rice production. Some varieties may have resistance to bird damage. The engagement of bird scarers and using cloth nets are commonly adopted for controlling birds in rice fields. Planting rice too early or late exposes the crop to severe bird damage, hence planting should be properly timed.

Striga hermonthica

Striga is increasingly becoming a serious problem in rice production in northern Nigeria and in Kano State, in particular, where the soils are sandy and major host plants for *Striga* such as maize and sorghum, are extensively cultivated. This weed is a parasite, which germinates only when the plants (hosts) which it attacks are present. *Striga* produces numerous tiny seeds, making it easy for the parasite to spread from place to place. Some of the control measures recommended for the control of *Striga* in rice include the use of *Striga*-tolerant rice varieties, use of *Striga*-free seeds, proper cleaning of farm tools especially after working on infested fields, proper fertilization (use of organic manure and inorganic fertilizer), crop rotation especially with soybean and groundnut, intercropping, and weeding. Rice farms should be weeded before *Striga* produces seeds to reduce the *Striga* seeds in the soil.



Plate 5. Rice field attacked by *Striga* plants.

Major diseases affecting rice fields and their control

Brown leaf spot

This is caused by the fungus *Helminthosporium oryzae*. This disease is widespread on rice in many northern states. The disease is seed-borne and can be transmitted to new crops through infected seed carried over from infected crop residues of the previous season, alternative crop and weed hosts, and contaminated irrigation water.

Burn all crop residues and alternative hosts, and use seeds from healthy plants. Apply Apron XL® at the rate of 10 g/5 kg seed before sowing. When symptoms are observed, spray Dithane M-45 weekly at the rate of 1.5 kg/ha (this is about 2–3 small matchboxes full per 15-L sprayer) for three weeks.

Blast

Blast disease is caused by the fungus, Pyricularia oryzae. It is very prevalent in lowland rice. It damages plants, especially during the seedling stage. It appears as neck rot and panicle blast after flowering. Cultural control methods include planting resistant varieties, avoiding close planting in the nursery, and adopting the recommended fertilization regime. For chemical control: apply Mancozeb or Dithane M-45 at 1–1.5 kg/ha when the sign of the disease is observed. Repeat application weekly for two weeks.



Plate 6. Rice field infected with blast disease

Harvesting

Rice is ready for harvesting when the grains are hard and turn yellow/brown (about 30–45 days after flowering). Rice is fully mature for harvesting when 80–85% of the grains are straw colored. To harvest, cut the rice stems with a sickle at about 10–15 cm above the ground. Tie the panicles in bundles, then place the tied-up bundles of harvested rice crop in an upright position for drying before threshing.

Threshing and winnowing

Threshing is done on hard surfaces by beating the grains out from the ears or through motorized threshers. Winnow to separate the chaff and empty grains from the well-filled mature grains. Dry the paddy to a safe moisture level of between 12 and 14% before storing. Freshly harvested paddy does not store well under high temperatures. Remove foreign matter in the paddy to avoid localized heating spots. Dry on mats spread on concrete floors slowly for the first few days to reduce breakage during milling. Avoid drying on bare floors to reduce the introduction of sand and pebbles and other foreign matter as they reduce the quality of the rice.



Plate 7. Motorized threshing of harvested rice.



Plate 8. Drying of threshed rice

Storage

Rice paddy intended for storage should be properly dried. Clean the storage container such as a rumbu before pouring in your paddy. To protect the paddy against insect pests, use 1½ matchboxes full of Coopex 2.5 to dust about 25 kg of paddy. Store in cool, dry, and rodent-proof conditions. Infested paddy should be fumigated with phostoxin in air-tight containers/rooms at the rate of one tablet/jute bag (100 kg paddy) or 10–15 tablets/ton of paddy.

Parboiling

All the varieties recommended in Table 1 are medium to long grain rice, and therefore need to be parboiled. Parboiling involves soaking the paddy in hot water at specified temperatures that vary with the varieties. For all the recommended varieties, parboiling



Plate 9. Storage structures used for paddy rice (a) Traditional and (b) Modern.

can be done by soaking paddy rice for 5–6 hours in hot water at 70 °C (hot enough for your fingers to withstand the temperature for about two seconds).

The paddy is later steamed for a short time (6–10 minutes) until the husks just open. Steaming dextrinizes the kernels and drives the vitamin thiamine and other water-soluble nutrients from the testa into the starchy core. After parboiling, dry under shade gradually to reduce breakage during milling.

Drying parboiled paddy

Drying: Parboiled paddy should be dried to 14% moisture content for safe storage or milling. Parboiled paddy is more difficult to dry and requires more energy than field paddy because its moisture content is much higher. However, higher air temperatures help reduce the drying time.



Plate 10. Threshed rice being parboiled.

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About Sasakawa Africa Association (SAA)

The Sasakawa Africa Association (SAA) began its work in Nigeria in 1992 when the Federal Ministry of Agriculture and Natural Resources (FMANR) and the SAA signed an MOU. Since then, SAA has been providing support to smallholder farmers throughout Nigeria with the full support of the government, focusing on **improving agricultural productivity**. It actively collaborates with National/States governments, Universities, international organizations and NGOs as well as farmer associations and private companies to promote agricultural extension activities. Since 2009, the activity has shifted from a primary focus on improving agricultural productivity to a **value chain approach** focused on market demand. While continuing to provide technical support to improve agricultural productivity and postharvest processing technology, SAA-Nigeria aims to contribute to improving the livelihood of farmers by selecting crops based on market demand. Under the new strategy (2021-2025), SAA-Nigeria will contribute to the establishment of a strong and sustainable food system in Nigeria by positioning **Regenerative Agriculture** as a core activity in response to climate change and soil degradation, promoting sustainable productivity improvement for smallholder farmers, and promoting a shift to **Market-oriented Agriculture** and **Nutrition-sensitive Agriculture**.

